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Missing Mandates for the standard library

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Introduction

As the "Mandating the standard library" papers have progressed/been applied, a few bits have been missed; mostly because parts of the library have moved.

This paper fixes those bits.

Thanks to Richard Smith for enumerating these parts.

Thanks to Tim Song for his review of R0.

Changes from R0:

- Removed the changes from [res.on.required]
- Fix a mis-edit in [coroutine.handle.observers]
- Add edits for [coroutine.handle.resumption]
- Add a missing edit for [util.smartptr.atomic.weak]/3

Wording

These changes are based on [N4842](#).

In [tab:cpp17 allocator], change the "Requires" to an "Preconditions":

~~Preconditions:~~~~Requires:~~ (**p*).m is well-defined.

~~Preconditions:~~~~Requires:~~ (**q*).m is well-defined.

~~Preconditions:~~~~Requires:~~ *p* ~~is~~~~shall be~~ a value returned by an earlier call to `allocate` that has not been invalidated by an intervening call to `deallocate`. *n* ~~matches~~~~shall match~~ the value passed to `allocate` to obtain this memory.

In [coroutine.handle.con]/2, change the "Requires" to an "Preconditions":

~~Preconditions:~~~~Requires:~~ *p* is a reference to a promise object of a coroutine.

In [coroutine.handle.export.import]/2, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ `addr` was obtained via a prior call to `address`

In [coroutine.handle.observers]/2, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ `*this` refers to a suspended coroutine

In [coroutine.handle.resumption], edit /2, /3 /4, as follows:

Preconditions:~~Requires:~~ `*this` refers to a suspended coroutine. The coroutine is not suspended at its final suspend point.

Effects: Resumes the execution of the coroutine. ~~If the coroutine was suspended at its final suspend point, behavior is undefined.~~

Preconditions:~~Requires:~~ `*this` refers to a suspended coroutine

In [coroutine.handle.promise]/1, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ `*this` refers to a coroutine.

In [util.smartptr.atomic.shared]/3, /6, and /12, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ order is neither~~The order argument shall not be~~ `memory_order::consume`, `memory_order::acquire`, nor `memory_order::acq_rel`.

Preconditions:~~Requires:~~ order is neither~~shall not be~~ `memory_order::release` nor `memory_order::acq_rel`.

Preconditions:~~Requires:~~ failure is neither~~shall not be~~ `memory_order::release` nor `memory_order::acq_rel`.

In [util.smartptr.atomic.weak]/3, /6 and /12, change the "Requires" to an "Preconditions":

Preconditions:~~Requires:~~ order is neither~~The order argument shall not be~~ `memory_order::consume`, `memory_order::acquire`, nor `memory_order::acq_rel`.

Preconditions:~~Requires:~~ order is neither~~shall not be~~ `memory_order::release` nor `memory_order::acq_rel`

Preconditions:~~Requires:~~ failure is neither~~shall not be~~ `memory_order::release` nor `memory_order::acq_rel`